PAPER_848: Chandra Sonification Collection — SNR 1181, H1821+643, IC 443, M74, MSH 15-52, SDSS J1531, Sgr A*

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Abstract

Eight systems from the Chandra Sonification Collection are analyzed. Sonification translates X-ray photon energies (0.5–7.0 keV) to audible frequencies (60–2000 Hz) on a logarithmic scale. H1821+643 is identified as the most massive cluster-central quasar known (~3–30 billion M_sun). Sgr A* exhibits negative buoyancy in the sonification context, confirming consistency across observational modalities.

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1. Systems and Parameters

System	M (kg)	r (m)	F_U_Bi (N)	Buoyancy
SNR 1181 (Pa 30)	2.5e30	2.47e19	~2.65e208	Positive
H1821+643	3e40	3.7e23	~2.65e208	Positive
Sonification Collection	1e35	3.09e20	~2.65e208	Positive
IC 443 (Jellyfish)	5.97e31	2.16e17	~2.65e208	Positive
M74 (Phantom Galaxy)	1.989e41	5.56e20	~2.65e208	Positive
MSH 15-52 (Hand PWN)	2.984e30	4.63e17	~2.65e208	Positive
SDSS J1531+3414	1.989e44	1.54e23	~2.65e208	Positive
Sgr A*	7.956e36	6.17e18	~-8.31e211	NEGATIVE

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2. Sonification Mapping

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X-ray energy -> audio frequency (logarithmic):

f_audio = f_min * (f_max / f_min)^((E - E_min) / (E_max - E_min))

E_min = 0.5 keV -> f_min = 60 Hz
E_max = 7.0 keV -> f_max = 2000 Hz

Example mappings:
  0.5 keV -> 60 Hz (deep bass)
  1.0 keV -> 127 Hz (low piano)
  2.0 keV -> 268 Hz (middle C region)
  4.0 keV -> 568 Hz (soprano range)
  7.0 keV -> 2000 Hz (bright treble)

Color mapping: red (low E) -> blue (high E)
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3. H1821+643 — Most Massive Cluster Quasar

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H1821+643: luminous quasar at center of massive galaxy cluster
Redshift: z ~ 0.299
M_BH ~ 3-30 billion M_sun (one of the most massive known)

M = 3e40 kg, r = 3.7e23 m

Despite extreme mass, large r keeps F_U_Bi positive.
The quasar's feedback is insufficient to trigger negative buoyancy
at cluster-scale radius.
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4. IC 443 and MSH 15-52

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IC 443 (Jellyfish Nebula): SNR interacting with molecular cloud
  M = 30 M_sun, r = 7 pc
  Molecular hydrogen shock heating visible in IR

MSH 15-52 (Hand PWN): pulsar wind nebula shaped like a hand
  PSR B1509-58: P = 150 ms, B ~ 1.5e13 G (near magnetar)
  Morphology: jet + equatorial torus + "fingers"
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5. Sgr A* in Sonification

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Sgr A*: M = 4e6 M_sun, r = 6.17e18 m
F_U_Bi ~ -8.31e211 N (NEGATIVE BUOYANCY)

Sonification of Sgr A* X-ray data produces distinctive
low-frequency signature from absorbed/scattered photons.
Negative buoyancy is consistent across all observational modalities.
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Conclusion

The 8-system Sonification Collection demonstrates that UQFF analysis is independent of observational presentation modality. X-ray to audio mapping provides an intuitive interface for non-visual data exploration. Sgr A*'s negative buoyancy persists across all analysis frameworks.
